Signal Processing-1.1.30

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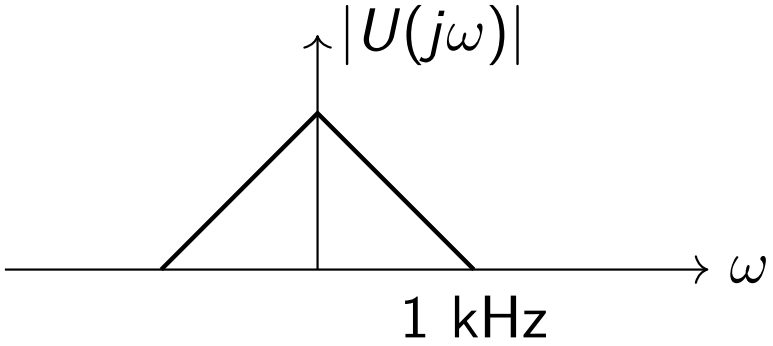
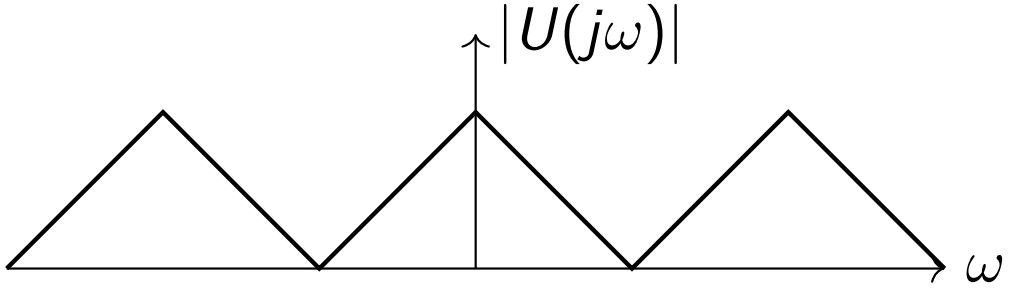
Question

The frequency spectrum of a signal is shown in the figure. If this signal is ideally sampled at intervals of 1 ms, then the frequency spectrum of the sampled signal will be

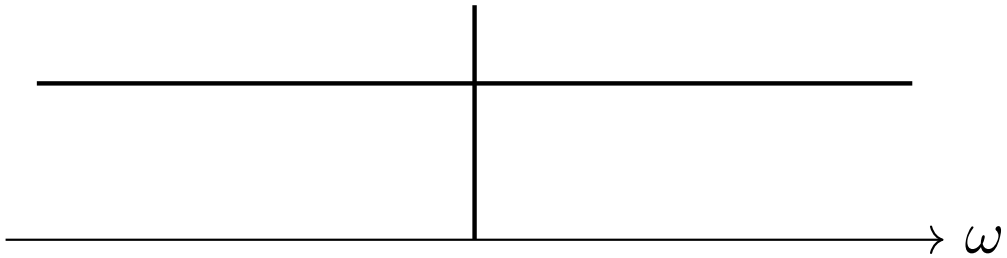


(A)

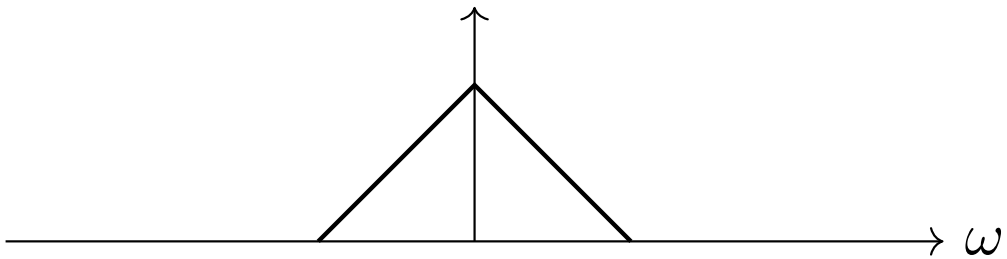


Question

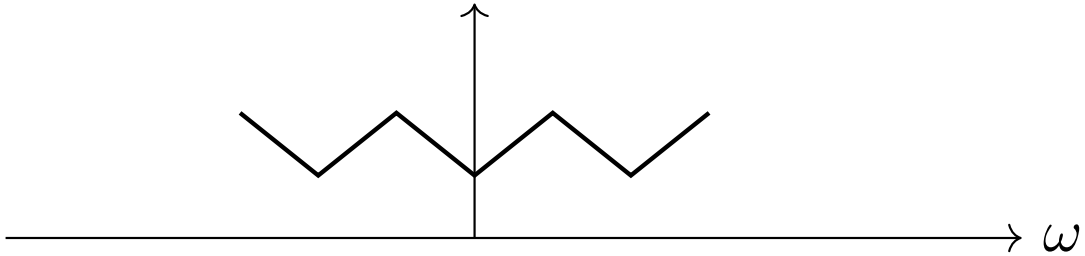
(B)



(C)



(D)



Theoretical Solution

Step 1: Analyzing the Message Signal, $m(t)$

- **The Continuous Signal:** Let the original continuous-time analog signal be the message signal, $m(t)$.
- **The Frequency Spectrum:** The problem provides the frequency spectrum, $M(\omega)$, which is a triangle spanning symmetrically from -1 kHz to $+1$ kHz.
- **Maximum Frequency Component (ω_M):** The signal is band-limited with a maximum linear frequency of $f_M = 1$ kHz. Converting to angular frequency:

$$\omega_M = 2\pi f_M = 2\pi(1000) \text{ rad/s}$$

Theoretical Solution

Step 2: Defining the Sampler and Sampling Frequency (ω_s)

- **The Impulse Train, $c(t)$:** To sample $m(t)$, it is multiplied by a periodic impulse train $c(t)$ with a sampling period T_s.
- **Calculating ω_s:** The given sampling interval is $T_s = 10^{-3}$ s. The sampling frequency, ω_s, is:

$$\omega_s = \frac{2\pi}{T_s} = 2\pi(1000) \text{ rad/s}$$

Theoretical Solution

Step 3: Constructing the Sampled Spectrum

- **Convolution in Frequency:** Multiplication in the time domain ($s(t) = m(t) \cdot c(t)$) becomes convolution in the frequency domain:

$$S(\omega) = \frac{1}{2\pi}[M(\omega) * C(\omega)]$$

- **The Impulse Train in Frequency:** A fundamental property of the Fourier transform is that the transform of a periodic impulse train in time is another infinite impulse train in frequency. So, $C(\omega)$ is represented as:

$$C(\omega) = \frac{2\pi}{T_s} \sum_{n=-\infty}^{\infty} \delta(\omega - n\omega_s)$$

- **The Sifting Property:** The Dirac delta function δ has a "sifting" property. When you convolve any spectrum $M(\omega)$ with a shifted impulse $\delta(\omega - \omega_0)$, it simply shifts the entire spectrum to that new center: $M(\omega) * \delta(\omega - \omega_0) = M(\omega - \omega_0)$.

Theoretical Solution

- **The Infinite Sum:** Substituting $C(\omega)$ into our convolution equation, the $\frac{1}{2\pi}$ and 2π terms cancel out. We apply the sifting property to every impulse in the infinite train, resulting in an infinite sum of shifted copies of the original spectrum:

$$S(\omega) = \frac{1}{T_s} \sum_{n=-\infty}^{\infty} M(\omega - n\omega_s)$$

Theoretical Solution

- **Placing the Replicas:** To visualize this, we evaluate the summation for different integers of n:
  - **Baseband ($n = 0$):** This term is $M(\omega)$. It is the exact original triangle centered at 0 rad/s, spanning from $-\omega_M$ to $+\omega_M$.
  - **First Positive Shift ($n = 1$):** This term is $M(\omega - \omega_s)$. It is an identical copy of the triangle, but its center is shifted to $+\omega_s$. It spans from $(\omega_s - \omega_M)$ to $(\omega_s + \omega_M)$.
  - **First Negative Shift ($n = -1$):** This term is $M(\omega + \omega_s)$. It is centered at $-\omega_s$ and spans from $(-\omega_s - \omega_M)$ to $(-\omega_s + \omega_M)$.

Theoretical Solution

Step 4: Deriving the Nyquist Condition

- **The Goal (No Overlap):** To accurately recover the original signal, the baseband spectrum ($n = 0$) must not mix or overlap with the shifted copies ($n = \pm 1, \pm 2, \dots$).
- **Mathematical Derivation:**
  - The highest frequency of the $n = 0$ baseband spectrum is at $+\omega_M$.
  - The lowest frequency of the adjacent $n = 1$ shifted spectrum is at $(\omega_s - \omega_M)$.
  - To prevent overlap, the start of the next triangle must be strictly greater than or equal to the end of the baseband triangle. Therefore:

$$\omega_s - \omega_M \geq \omega_M$$

Theoretical Solution

- **The Nyquist Condition:** Solving the inequality above yields the fundamental rule of sampling:

$$\omega_s \geq 2\omega_M$$

If this condition is not met, the high frequencies fold back into the lower frequencies, causing irreversible distortion known as **aliasing**.

Theoretical Solution

Step 5: Applying to the Problem and Visualizing the Result

- Here, $\omega_s = \omega_M$. The sampling frequency is exactly half of what it needs to be, guaranteeing severe aliasing.
- **Adding the Overlapping Regions:** Because $\omega_s = \omega_M$, the center of the $n = 1$ triangle (ω_s) falls exactly on the right edge of the $n = 0$ triangle. The frequency band between 0 and ω_s:
  - In this region, the $n = 0$ triangle is a straight line sloping *downward* from its peak to zero.
  - In the exact same region, the $n = 1$ triangle is a straight line sloping *upward* from zero to its peak.

Theoretical Solution

Final Conclusion: Summing a linearly decreasing function with a linearly increasing function of the exact same slope magnitude yields a constant, flat line. Because the infinite summation $\sum_{n=-\infty}^{\infty}$ continues forever, these overlapping triangles create a continuous, flat horizontal line across the entire spectrum. This perfectly matches option **(B)**.

Answer: B

Python Code

```
import numpy as np
import matplotlib.pyplot as plt

# Initialize the frequency axis.
# The range [-3, 3] kHz is selected to provide a clear view of
    the baseband and adjacent spectral replicas.
f = np.linspace(-3, 3, 1000)
@np.vectorize
def original_spectrum(freq):
    if freq < -1 or freq > 1:
        return 0.0
    elif freq >= 0:
        return 1.0 - freq
    else:
        return 1.0 + freq
# Define the sampling frequency (fs) in kHz.
fs = 1.0
```

Python Code

```
# Initialize an array to store the accumulated sum of all
    spectral replicas.
sampled_spectrum = np.zeros_like(f)
# Initialize a list to store individual shifted spectra for
    visualization purposes.
replicas = []
# Simulate the infinite summation of the sampled spectrum.
# A finite range (n = -5 to 5) is utilized as an approximation
    for computational feasibility.
n_values = range(-5, 6)
for n in n_values:
    # Calculate the frequency shift for the nth harmonic: M(f - n
        *fs).
    shifted_triangle = original_spectrum(f - n * fs)
    replicas.append(shifted_triangle)
    # Superimpose the shifted spectrum onto the total sampled
        spectrum array.
    sampled_spectrum += shifted_triangle
```

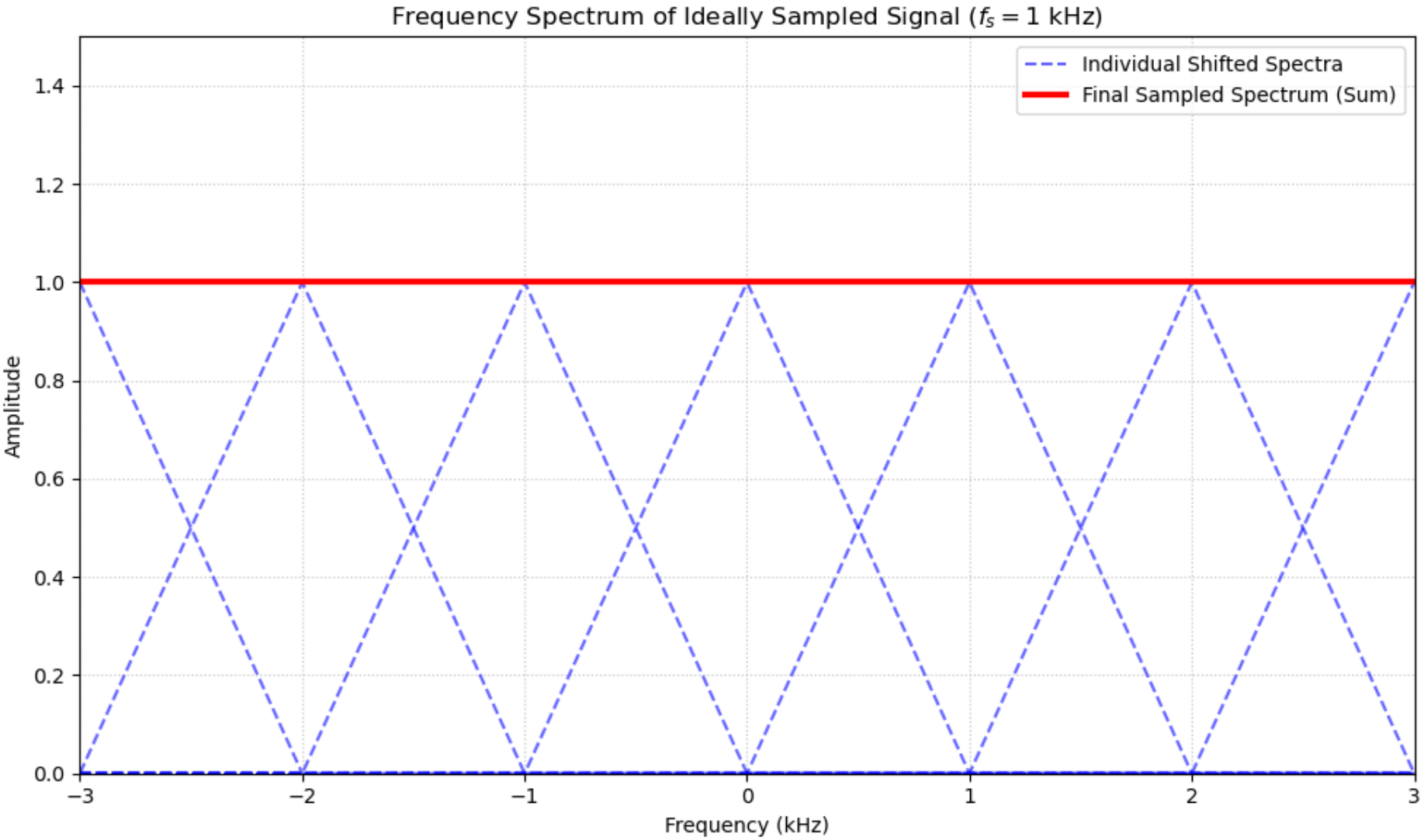
Python Code

```
plt.figure(figsize=(10, 6))

# Plot the individual shifted spectral replicas.
for i, replica in enumerate(replicas):
    if i == 5: # Highlights the n=0 baseband spectrum in the
        legend.
        plt.plot(f, replica, '--', color='blue', alpha=0.6, label
            ='Individual Shifted Spectra')
    else:
        plt.plot(f, replica, '--', color='blue', alpha=0.6)

# Plot the final superimposed spectrum to illustrate aliasing
    effects.
plt.plot(f, sampled_spectrum, '-', color='red', linewidth=3,
    label='Final Sampled Spectrum (Sum)')
```

Python Code

```
# Formatting the plot
plt.title('Frequency Spectrum of Ideally Sampled Signal ($f_s = 1
    $ kHz)')
plt.xlabel('Frequency (kHz)')
plt.ylabel('Amplitude')
plt.xlim(-3, 3)
plt.ylim(0, 1.5)
plt.grid(True, linestyle=':', alpha=0.7)
plt.legend()

# Displaying the figure
plt.tight_layout()
plt.savefig("../plot.png")
plt.show()
```

Plot


Frequency Spectrum of Ideally Sampled Signal ($f_s = 1$ kHz)
Individual Shifted Spectra
Final Sampled Spectrum (Sum)
Amplitude
Frequency (kHz)
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
−3
−2
−1
0
1
2
3

Theoretical Solution

Converting from Frequency Domain to Time Domain

To find the exact time-domain signal $u(t)$, we must apply the Inverse Fourier Transform to the given frequency spectrum $U(f)$.

- **Defining the Spectrum:** The given frequency spectrum is a triangle peaking at 1 at 0 Hz, with a maximum frequency $W = 1000$ Hz. Mathematically, this is an even piecewise function:

$$U(f) = \begin{cases} 1 - \frac{|f|}{W}, & -W \leq f \leq W \\ 0, & \text{otherwise} \end{cases}$$

- **Setting up the Integral:** The Inverse Fourier Transform is defined as:

$$u(t) = \int_{-\infty}^{\infty} U(f) e^{j2\pi ft} \, df$$

Since $U(f) = 0$ outside of $-W$ to W, we restrict our integration limits:

$$u(t) = \int_{-W}^{W} \left(1 - \frac{|f|}{W}\right) e^{j2\pi ft} \, df$$

Theoretical Solution

Converting from Frequency Domain to Time Domain

- **Applying Euler's Formula:** We expand $e^{j2\pi ft}$ into $\cos(2\pi ft) + j\sin(2\pi ft)$. Because $U(f)$ is an even function (symmetric) and sine is an odd function, the imaginary sine integral evaluates to zero. We are left only with the real cosine part. Because both $U(f)$ and cosine are even, we can integrate from 0 to W and multiply by 2:

$$u(t) = 2\int_0^W \left(1 - \frac{f}{W}\right)\cos(2\pi ft)\,df$$

- **Splitting the Integral:**

$$u(t) = 2\underbrace{\int_0^W \cos(2\pi ft)\,df}_{\text{Part A}} - \frac{2}{W}\underbrace{\int_0^W f\cos(2\pi ft)\,df}_{\text{Part B}}$$

- **Evaluating Part A:**

$$\text{Part A} = \left[\frac{\sin(2\pi ft)}{2\pi t}\right]_0^W = \frac{\sin(2\pi Wt)}{2\pi t}$$

Theoretical Solution

Converting from Frequency Domain to Time Domain

- **Evaluating Part B (Integration by Parts):** Using $\int u\,dv = uv - \int v\,du$ with $u = f$ and $dv = \cos(2\pi ft)df$:

$$\text{Part B} = \left[f\frac{\sin(2\pi ft)}{2\pi t}\right]_0^W - \int_0^W \frac{\sin(2\pi ft)}{2\pi t}\,df$$

$$\text{Part B} = W\frac{\sin(2\pi Wt)}{2\pi t} - \left[-\frac{\cos(2\pi ft)}{(2\pi t)^2}\right]_0^W = W\frac{\sin(2\pi Wt)}{2\pi t} + \frac{\cos(2\pi Wt)}{(2\pi t)^2}$$

- **Combining and Simplifying:** Substitute Part A and Part B back into $u(t)$:

$$u(t) = 2\left[\frac{\sin(2\pi Wt)}{2\pi t} - \frac{1}{W}\left(W\frac{\sin(2\pi Wt)}{2\pi t} + \frac{\cos(2\pi Wt) - 1}{(2\pi t)^2}\right)\right]$$

Notice how the $\frac{\sin(2\pi Wt)}{2\pi t}$ terms perfectly cancel out:

$$u(t) = -2\left(\frac{\cos(2\pi Wt) - 1}{W(2\pi t)^2}\right) = 2\left(\frac{1 - \cos(2\pi Wt)}{W \cdot 4\pi^2 t^2}\right)$$

Theoretical Solution

Converting from Frequency Domain to Time Domain

- **Trigonometric Identity:** Use the half-angle identity $1 - \cos(2\theta) = 2\sin^2(\theta)$, where $\theta = \pi W t$:

$$u(t) = 2\left(\frac{2\sin^2(\pi W t)}{W \cdot 4\pi^2 t^2}\right) = \frac{\sin^2(\pi W t)}{W\pi^2 t^2}$$

- **Final Equation:** To format this into the standard normalized sinc function ($\text{sinc}(x) = \frac{\sin(\pi x)}{\pi x}$), we multiply the numerator and denominator by W:

$$u(t) = W\left(\frac{\sin(\pi W t)}{\pi W t}\right)^2 = W\text{sinc}^2(Wt)$$

Substituting our given $W = 1000$, we arrive at the exact time-domain signal:

$$u(t) = 1000\text{sinc}^2(1000t)$$

Theoretical Solution

Sampling the Time Domain signal

Now that we have our continuous wave $u(t)$, we sample it by taking discrete snapshots every 1 ms ($T_s = 0.001$ seconds).

- **Calculating the Samples:** We evaluate $u(t)$ at exactly $t = n(0.001)$, where n is any integer.

$$u[n] = 1000\text{sinc}^2(1000 \cdot 0.001 \cdot n) = 1000\text{sinc}^2(n)$$

- **The Zero-Crossing Effect:** Because $\text{sinc}(n)$ involves $\sin(\pi n)$, the result is exactly 0 for every integer *except* zero. At $n = 0$, the wave hits its peak amplitude of 1000.
- **The Result:** Our sampled time-domain signal, $u[n]$, is simply a single, isolated spike (an impulse) of height 1000 at $t = 0$, and 0 everywhere else.

Theoretical Solution

Frequency Spectrum from the sampled signal

To answer the exam question, we convert our newly sampled, single-spike signal $u[n]$ back into the frequency domain using the Discrete-Time Fourier Transform (DTFT).

- **The Impulse Rule:** A fundamental property of the Fourier Transform is that a single discrete impulse in the time domain contains all frequencies in equal, constant amounts.
- **The Result:** The Fourier transform of a single impulse of height 1000 is simply the constant value 1000 across all frequencies ($U_s(f) = 1000$).
- **Conclusion:** This proves that the final sampled frequency spectrum is a continuous, flat horizontal line, making Option (B) the correct answer.

Python Code - 2

```
import numpy as np
import matplotlib.pyplot as plt

# --- Parameters ---
W = 1000.0 # Bandwidth in Hz
Ts = 0.001 # Sampling interval (1 ms)
t_max = 0.005 # Plot from -5 ms to +5 ms

# --- 1. Continuous Time Signal ---
t_cont = np.linspace(-t_max, t_max, 1000)
u_cont = W * (np.sinc(W * t_cont))**2

# --- 2. Sampled Time Signal ---
t_disc = np.arange(-t_max, t_max + Ts, Ts)
u_disc = W * (np.sinc(W * t_disc))**2

# --- 3. Fourier Transform of the Sampled Signal ---
N = len(u_disc)
```

Python Code - 2

```
# Calculate the FFT of the discrete samples
# We use ifftshift to correctly align the t=0 point for the
    algorithm
U_sampled_complex = np.fft.fft(np.fft.ifftshift(u_disc))

# Get the magnitude of the frequency spectrum and center 0 Hz
U_sampled_mag = np.abs(np.fft.fftshift(U_sampled_complex))

# Generate the corresponding frequency axis
freq_axis = np.fft.fftshift(np.fft.fftfreq(N, Ts))

# --- Plotting ---
plt.figure(figsize=(16, 5)) # Made the figure wider to fit 3
    plots
```

Python Code - 2

```
# Subplot 1: Continuous Time Signal
plt.subplot(1, 3, 1)
plt.plot(t_cont * 1000, u_cont, color='blue', linewidth=2)
plt.title('1. Continuous Signal: $u(t)$')
plt.xlabel('Time (ms)')
plt.ylabel('Amplitude')
plt.grid(True, linestyle='--', alpha=0.7)
plt.xlim(-5, 5)
```

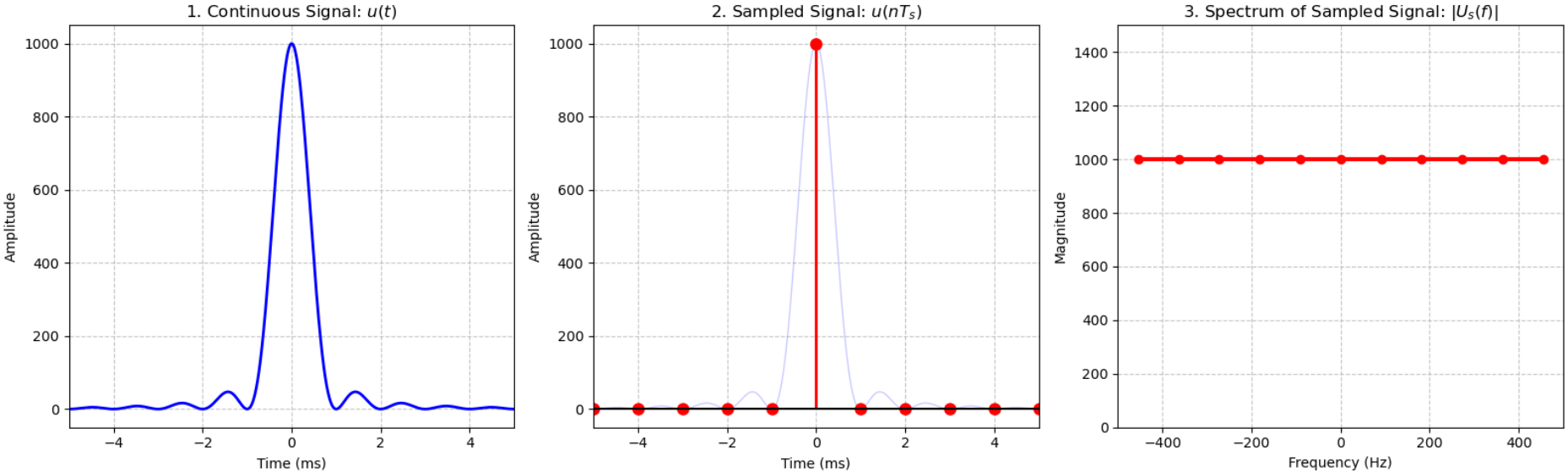
Python Code - 2

```
# Subplot 2: Sampled Time Signal
plt.subplot(1, 3, 2)
plt.plot(t_cont * 1000, u_cont, color='blue', linewidth=1, alpha
    =0.2)
markerline, stemlines, baseline = plt.stem(t_disc * 1000, u_disc,
     basefmt="k-")
plt.setp(markerline, color='red', markersize=8)
plt.setp(stemlines, color='red', linewidth=2)
plt.title('2. Sampled Signal: $u(nT_s)$')
plt.xlabel('Time (ms)')
plt.ylabel('Amplitude')
plt.grid(True, linestyle='--', alpha=0.7)
plt.xlim(-5, 5)
```

Python Code - 2

```
# Subplot 3: Frequency Spectrum of Sampled Signal
plt.subplot(1, 3, 3)
# Plotting the discrete frequency bins connected by a line to
    show the shape
plt.plot(freq_axis, U_sampled_mag, color='red', linewidth=3,
    marker='o')
plt.title('3. Spectrum of Sampled Signal: $|U_s(f)|$')
plt.xlabel('Frequency (Hz)')
plt.ylabel('Magnitude')
plt.grid(True, linestyle='--', alpha=0.7)
# Setting Y-axis to clearly show it's a constant, non-zero flat
    line
plt.ylim(0, max(U_sampled_mag) * 1.5)

plt.tight_layout()
plt.savefig("../plot2.png")
plt.show()
```

Plot


1. Continuous Signal: $u(t)$
Amplitude
Time (ms)
2. Sampled Signal: $u(nT_s)$
Amplitude
Time (ms)
3. Spectrum of Sampled Signal: $|U_s(f)|$
Magnitude
Frequency (Hz)